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6. Worked example: variation of parameters

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Steps to solve $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r}$ by variation of parameters:

1. Find a fundamental matrix $\mathbf{X}$ of the associated homogeneous system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$. One way to do this is to use eigenvalues and eigenvectors of $\mathbf{A}$ to find a basis of homogeneous solutions and use these solutions as the columns of $\mathbf{X}$.
2. Substitute $\mathbf{x} = \mathbf{X}\mathbf{v}$ for a vector-valued function $\mathbf{v}$; this eventually leads to

$$\dot{\mathbf{v}} = \mathbf{X}^{-1}\mathbf{r}.$$

(and you may jump right to this if you want).

3. Compute the right hand side and integrate each component function to find $\mathbf{v}$:

$$\mathbf{v}(t) = \int \mathbf{X}^{-1}\mathbf{r} dt = \mathbf{v}_p + \mathbf{c}. \tag{4.135}$$

Note that the indefinite integral will contain a constant of integration $+\mathbf{c}$.

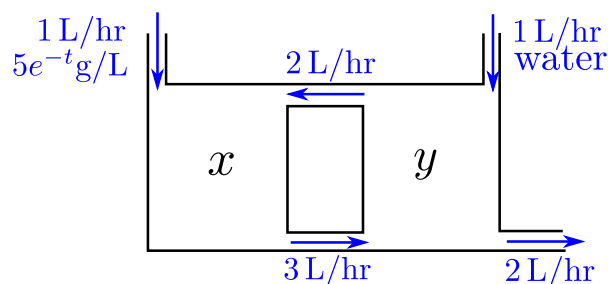
4. The general solution to the inhomogeneous equation is

$$\mathbf{x} = \mathbf{X}\mathbf{v} \tag{4.136}$$

$$= \mathbf{X}\mathbf{v}_p + \mathbf{X}\mathbf{c} \tag{4.137}$$

$$= \mathbf{x}_p + \mathbf{x}_h. \tag{4.138}$$

Example 6.1 Let us finish the example from the video.



Two connected 1-liter tanks with salt solution.

Let x (in grams) and y (in grams) be the amounts of salt in the left and right tanks respectively. The salt solution flows through the bottom pipe to right at **3 liters/hour**, and through the top pipe to the left at **2 liters/hour**.

A salt solution of concentration **$5e^{-t}$ grams/liter** is mixed into the system through a pipe connected to the left tank at a flow rate of **1 liters/hour**. In addition, water flows into the right tank through a top pipe at the rate of **1 liters/hour**, and the salt solution inside the right tank flows out through a bottom pipe at the rate of **2 liters/hour**.

Our goal is to find how the concentration of salt in each tank evolves over time.

Recall that the inhomogeneous system of ODEs that describes the rate of change of the amount of salt in the tanks is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r}(t) \quad \text{where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \tag{4.139}$$

$$\mathbf{A} = \begin{pmatrix} -3 & 2 \\ 3 & -4 \end{pmatrix} \tag{4.140}$$

$$\mathbf{r} = \begin{pmatrix} 5e^{-t} \\ 0 \end{pmatrix}. \tag{4.141}$$

To use variation of parameters, we first find a fundamental matrix of the associated homogeneous system.

To use variation of parameters, we first find a fundamental matrix of the associated homogeneous system. Recall (from a previous problem) that a fundamental matrix, whose columns are the normal modes, is given by the product

$$\mathbf{X} = \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix}.$$

(4.142)

Then, variation of parameters gives the general solution

$$\mathbf{x} = \mathbf{X}\mathbf{v} \quad \text{where } \mathbf{v} = \int \mathbf{X}^{-1}\mathbf{r} \, dt.$$

(4.143)

Let us find $\mathbf{v}$. First, compute $\mathbf{X}^{-1}\mathbf{r}$ (we show our work but you can do this in different ways):

$$\mathbf{X}^{-1} = \left(\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix} \right)^{-1}$$

(4.144)

$$= \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix}^{-1} \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} e^t & 0 \\ 0 & e^{6t} \end{pmatrix} \left(\frac{1}{5} \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \right)$$

(4.145)

$$\mathbf{X}^{-1}\mathbf{r} = \frac{1}{5} \begin{pmatrix} e^t & 0 \\ 0 & e^{6t} \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5e^{-t} \\ 0 \end{pmatrix} = \begin{pmatrix} e^t & 0 \\ 0 & e^{6t} \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{-t} \\ 0 \end{pmatrix}$$

(4.146)

$$= \begin{pmatrix} e^t & 0 \\ 0 & e^{6t} \end{pmatrix} \begin{pmatrix} 3e^{-t} \\ -e^{-t} \end{pmatrix} = \begin{pmatrix} 3 \\ -e^{5t} \end{pmatrix}.$$

(4.147)

Therefore,

$$\mathbf{v}(t) = \int \mathbf{X}^{-1}\mathbf{r} \, dt = \int \begin{pmatrix} 3 \\ -e^{5t} \end{pmatrix} dt = \begin{pmatrix} \int 3 \, dt \\ \int -e^{5t} \, dt \end{pmatrix} = \begin{pmatrix} 3t \\ -\frac{1}{5}e^{5t} \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

(4.148)

This gives the general solution to the inhomogeneous system:

$$\mathbf{x} = \mathbf{X}\mathbf{v} \quad \text{where } \mathbf{X} = \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3t \\ -\frac{1}{5}e^{5t} \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

(4.149)

$$= \underbrace{\mathbf{X} \begin{pmatrix} 3t \\ -\frac{1}{5}e^{5t} \end{pmatrix}}_{\text{particular solution}} + \underbrace{\mathbf{X} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\text{general homogeneous solution}}$$

(4.150)

$$= \underbrace{\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix} \begin{pmatrix} 3t \\ -\frac{1}{5}e^{5t} \end{pmatrix}}_{\text{particular solution}} + \underbrace{\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-6t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\text{general homogeneous solution}}$$

(4.151)

$$= \underbrace{\begin{pmatrix} 2 \\ -3 \end{pmatrix} \frac{e^{-t}}{5} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} 3te^{-t}}_{\text{particular solution}} + \underbrace{c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} -2 \\ 3 \end{pmatrix} e^{-6t}}_{\text{general homogeneous solution}}.$$

(4.152)

This decays to zero as $t \rightarrow \infty$, so the amount of salt in both tanks will eventually become negligible. For fun, find the maximum salt concentration in each tank. Also explore how the different input concentration will affect the long term behavior.

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